

LITERATURE REVIEW TABLE

Comparative analysis of 29 supplied sources against NEW INDIA HEALTHCARE DATASET 2026

Dataset Profile	Value
Records	100,000
Features	39
States/UTs represented	10
Primary diagnoses	10
Age range	27–86 years
Visit period	2024-05-27 to 2026-05-26
Hospital types	3
Treatment types	7
Treatment outcomes	5
Imaging types	5
Glycemic classes	3

Core dataset advantage used for gap analysis: 100,000 patient records, 39 features, 10 primary diagnosis categories, 10 states/UTs, demographics, socioeconomic status, occupation, visit time, symptoms and binary symptom indicators, fasting glucose, glycemic status, HbA1c, cholesterol, BMI, treatment type/outcome, imaging, hospital type, and insurance coverage.

S. No.	Author(s) & Year	Title of the Paper	Dataset	Method / Algorithm	Key Features / Input	Results / Accuracy	Limitations	Research Gap vs. NEW INDIA HEALTHCARE DATASET 2026
1	Butt et al., 2021	Machine Learning Based Diabetes Classification and Prediction for Healthcare Applications	Multiple diabetes datasets; ML/IoT context	ML classifiers; comparative prediction	Clinical + glucose variables	Comparative diabetes prediction; no single universal accuracy emphasized	Primarily diabetes-focused; heterogeneous benchmark data	Broader India-specific multi-disease, treatment, imaging, and socioeconomic data needed.
2	Raja, 2024	Artificial Intelligence-Based Healthcare Data Analysis Using MPNN Based on Optimal Feature Selection	Diabetes dataset; author-provided/on-request	SSOSSV feature selection + MPNN; AES/blockchain	Selected diabetic clinical features	94.46% accuracy reported	Single-disease focus; dataset not openly specified	Test feature selection on 100k Indian records with 10 diagnoses and richer care context.
3	Authors, 2025	A Survey of Machine Learning Techniques for Diabetes Prediction: Current Trends and Future Directions	Reviewed diabetes datasets, commonly Pima/UCI and related sources	Systematic survey of ML/DL techniques	Diabetes risk/clinical variables	Survey; no single accuracy	Fragmented datasets; limited external validation and generalizability	Use one large, heterogeneous Indian dataset for comparative multi-model benchmarking.
4	Srivastava et al., 2024/25	Complex Predictive Analysis for Health Care: A Comprehensive Review	Literature-based review	Predictive analytics/ML review	Healthcare predictive variables across studies	Review; no single accuracy	Conceptual breadth; no common large Indian benchmark	Empirical validation on a 100,000-record India-focused dataset.
5	Authors, 2025	IEEE document 11377214 (publisher metadata not reliably retrievable)	NR from accessible abstract metadata	NR	NR	NR	Direct abstract could not be reliably retrieved	Retain as source-mapped evidence only; validate against accessible publisher PDF before final thesis submission.
6	Authors, 2024	Taylor & Francis article, DOI: 10.1080/03772063.2024.2434572	NR from accessible abstract metadata	NR	NR	NR	Abstract metadata inaccessible in automated retrieval	Requires source-level verification; compare with India dataset only after full-text extraction.
7	Site, Nurmi & Lohan, 2021	Systematic Review on Machine-Learning Algorithms Used in Wearable-Based eHealth Data Analysis	Wearable/eHealth datasets reviewed	Systematic review of ML algorithms	Physiological and sensor time-series data	Review; no unified accuracy	Device heterogeneity, privacy, and limited interoperability	India dataset adds clinical, socioeconomic and hospital variables beyond wearables.
8	Abdullah et al., 2024	Disseminating the Risk Factors With Enhancement in Precision Medicine Using Comparative Machine Learning Models for Healthcare Data	Healthcare datasets for comparative ML	Comparative ML for risk-factor analysis	Risk factors and patient attributes	Comparative performance study	Population/context transferability remains limited	Evaluate precision-medicine risk models on geographically and socially diverse Indian records.
9	Authors, 2026	A Hybrid Machine Learning Algorithm for Clustering Healthcare Data Improving Prediction Accuracy	Healthcare clustering/prediction datasets	Hybrid ML clustering + prediction	Mixed healthcare attributes	Improved prediction claimed; task-specific metrics	Generalization across health systems and mixed real-world data remains challenging	Use 39-feature Indian dataset for mixed-feature clustering and downstream diagnosis prediction.
10	Authors, 2022	A Review Paper on Machine Learning Techniques and Its Applications in Health Care Sector	Reviewed healthcare datasets	Review of supervised/unsupervised ML	Clinical, diagnostic and administrative features	Review; no single accuracy	No unified benchmark or India-specific validation	Provide a large common benchmark spanning diagnosis, treatment, imaging and social context.
11	Authors, 2022	IEEE document 9694592 (publisher metadata not reliably retrievable)	NR	NR	NR	NR	Publisher abstract unavailable to automated retrieval	Needs full-text verification before quantitative comparison.
12	Authors, 2025/26	IEEE document 11646107 (publisher metadata not reliably retrievable)	NR	NR	NR	NR	Publisher abstract unavailable to automated retrieval	Needs full-text verification before quantitative comparison.
13	Authors, NR	ProQuest source c5767a1e3c1c9b08b6612ebff536d30b	Access-restricted / metadata not exposed	NR	NR	NR	Institutional access restriction prevented reliable extraction	Do not infer findings; retain as a source requiring manual full-text verification.
14	Khekare et al., 2025	Advanced Fuzzy Logic for Predicting Hospital Readmissions in Diabetic Patients	Diabetes/readmission data discussed; pilot clinical prediction context	Fuzzy logic + predictive modeling / LSTM-related methods	Readmission and diabetic clinical variables	Pilot accuracy reported around 83% in accessible metadata	Readmission-specific; deployment and long-term validation still needed	New dataset supports outcome-aware models using treatment outcome, hospital type and insurance.
15	Nada et al., NR	Kingston repository PDF: Nada-P-51193-VoR	PDF metadata not reliably retrievable from supplied temporary URL	NR	NR	NR	Temporary/tokenized link could not be reliably analyzed	Requires stable PDF or DOI for defensible extraction.
16	Authors, 2023	IEEE document 10266089 (publisher metadata not reliably retrievable)	NR	NR	NR	NR	Abstract not reliably retrievable	Requires publisher/full-text verification.
17	Authors, 2023	IEEE document 10075459 (publisher metadata not reliably retrievable)	NR	NR	NR	NR	Abstract not reliably retrievable	Requires publisher/full-text verification.
18	Authors, 2023/24	IEEE document 10370452 (publisher metadata not reliably retrievable)	NR	NR	NR	NR	Abstract not reliably retrievable	Requires publisher/full-text verification.
19	Authors, 2024	Local file: jcssp.2024.1146.1165.pdf	Not uploaded; local file:// path is not web-accessible	NR	NR	NR	The referenced PDF is unavailable outside the original computer	Upload the PDF to complete a defensible row-level analysis.
20	Authors, 2015/16	ScienceDirect PII S0020025515007288	Information Sciences article; metadata not fully exposed by source access	NR	NR	NR	Publisher page access restricted in automated retrieval	Full abstract/full text needed before exact method/result comparison.
21	Authors, 2016/17	IEEE document 7813730 (publisher metadata not reliably retrievable)	NR	NR	NR	NR	Robots restriction prevented direct abstract extraction	Requires full-text verification before quantitative use.
22	Ullah & Garcia-Zapirain, 2024	Quantum Machine Learning Revolution in Healthcare: A Systematic Review of Emerging Perspectives and Applications	49 selected healthcare/QML studies after screening	Systematic review of quantum ML	Healthcare datasets and QML applications	Review; no clinical benchmark accuracy	Qubit noise, scalability, error correction and immature clinical validation	Conventional large Indian benchmark can establish strong classical baselines before QML experiments.
23	Authors, 2023	Taylor & Francis article, DOI: 10.1080/21681163.2023.2264937	Publisher-access restricted during automated retrieval	NR	NR	NR	Full abstract unavailable from supplied page	Verify title, dataset and metrics from full text before final quantitative citation.
24	Jackins et al., 2021	AI-Based Smart Prediction of Clinical Disease Using Random Forest Classifier and Naive Bayes	Pima diabetes, Framingham heart disease, Wisconsin breast cancer	Random Forest vs Gaussian Naive Bayes	Clinical/demographic disease variables	Diabetes test: NB 74.46%, RF 74.03%; heart: RF 83.85%; cancer: RF 92.40%	Benchmark datasets are population-specific and task-separated	Single India dataset enables unified multi-disease and outcome modeling.
25	Pati et al., 2022	Diagnose Diabetic Mellitus Illness Based on IoT Smart Architecture	Pima Indians + Frankfurt Hospital diabetes datasets and integrated data	Ensemble DL with IoT–fog–cloud architecture	Diabetes clinical features	Multi-metric evaluation; architecture-focused	Diabetes-only and infrastructure emphasis; limited India-wide heterogeneity	Evaluate scalable clinical prediction using 10 diagnoses plus treatment, imaging and hospital context.
26	Maurya et al., 2025	Federated Learning for Privacy-Preserving Severity Classification in Healthcare: A Secure Edge-Aggregated Approach	Healthcare severity-classification data	Federated learning; secure edge aggregation	Distributed patient/clinical features	Improved convergence and reliability reported	Federated deployment still depends on site heterogeneity and data governance	Partition India dataset by state/hospital type to simulate realistic federated heterogeneity.
27	Authors, 2018/19	A PHR-Based System for Monitoring Diabetes in Mobile Environment	Personal health record/mobile diabetes data context	Mobile PHR monitoring system	Patient-entered/diabetes monitoring variables	System/application evaluation rather than broad multi-disease benchmark	Diabetes-specific; mobile self-monitoring focus	Extend from diabetes monitoring to multi-disease risk and treatment-outcome decision support.
28	Almadani et al., 2025	A Systematic Survey of Distributed Decision Support Systems in Healthcare	PRISMA review of DDSS literature	Distributed decision support; AI, IoMT, blockchain	Clinical decision, integration and distributed data	Survey; no unified accuracy	Security, privacy, interoperability, data standardization and alert fatigue	New dataset supports testing an interpretable decision-support pipeline with hospital and insurance context.
29	Authors, 2019	MDPI Electronics article 8(7):768	Publisher page supplied; metadata not reliably retrievable in automated access	NR	NR	NR	Automated publisher retrieval unavailable	Full-text verification required before exact dataset/method/result claims.

SOURCE ACCESS AND VALIDATION NOTE

Rows were created from the 29 supplied links and the uploaded NEW INDIA HEALTHCARE DATA 2026.xlsx. Where publisher pages exposed reliable title/abstract/metadata, the table records specific dataset, method and reported results. Where access restrictions, robots restrictions, institutional login, temporary tokenized URLs, or a local file:// path prevented reliable extraction, the row is explicitly marked NR or requires full-text verification rather than inventing findings. The strongest directly supported research gap across the accessible literature is the lack of a single large, heterogeneous, India-focused dataset combining multi-disease diagnosis with clinical biomarkers, symptoms, socioeconomic variables, hospital context, treatment, treatment outcome, imaging and insurance information.

Recommended thesis-level research direction: Develop and compare interpretable ML, ensemble ML, deep learning and privacy-preserving/federated approaches for (1) multi-class primary diagnosis prediction, (2) glycemic-status prediction, and (3) treatment-outcome prediction; perform state/hospital-type subgroup validation and report accuracy, macro-F1, recall, AUROC where applicable, calibration, class-wise error, and fairness across gender, age group and socioeconomic status.